

THE UNIVERSITY *of* EDINBURGH
School of Physics
and Astronomy

Senior Honours Project

Watching Coffee Dry

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Abstract

In this project, the ratio of coffee ring diameter to initial contact angle was measured for a colloidal solution resting on a range of substrates using a drop shape analyser and optical microscope setup. A negative correlation between this ratio and the initial contact angle was observed qualitatively using image analysis techniques with the Fiji (ImageJ) software package.

Declaration: I declare that this project and report is my own work.

Signature: 

Date: March 23, 2026

Supervisor: Job Thijssen

Duration: 10 Weeks

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1 Introduction

When a colloid is left to dry on a surface one may observe a collection of particles in its wake. This is called the coffee ring effect. This occurs due to an inhomogeneity in a droplets shape when the three-phase contact line is pinned to a surface and the droplet is left to dry. The perimeter of the droplet has more surface area exposed to the environment and so evaporates faster than the top of the droplet. This evaporation causes a capillary flow of fluid towards the perimeter of the droplet where a collection of particles forms after the droplet has evaporated completely [1, 2]. This effect has profound impact in the inkjet printing industry because inks are amongst the many consumer products which rely on colloidal particle dispersions to operate [3, 4].

It has been almost 2 centuries since Thomas Young originally posited the relationship between contact angle (see Figure 1) and interfacial tensions for a three-phase system [5], however, modern literature seems devoid of research which directly correlates the coffee ring particle deposition and initial contact angle of a solution on a substrate.

The aim of this project is to verify qualitatively whether or not a relationship between initial contact angle on a substrate and the coffee ring particle deposition exists. To verify this, a colloidal silica solution was prepared and dropped onto substrates with varying hydrophobicities to sample an extensive range of contact angles [6]. The initial contact angle was measured with a Krüss DSA100 drop shape analyser and then the droplets were left to dry before being imaged by an Olympus BX50 optical microscope. After analysis with the ImageJ image analysis software, the particle deposition was characterised by the unitless quantity:

$$R(\theta) = \frac{\phi_{outer}}{\phi_{outer} - \phi_{inner}} \tag{1}$$

where θ is the contact angle for the substrate and ϕ represents diameter measurements of the coffee ring. This quantity was chosen because it is independent of both volume and length scale.

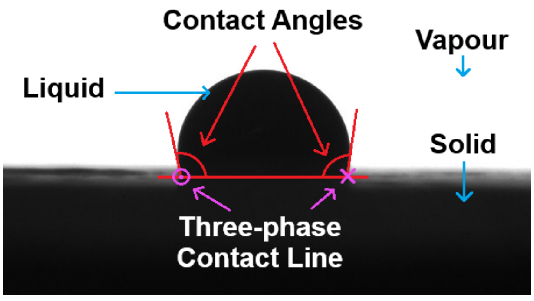


Figure 1: **Image of a droplet resting on a substrate produced by the Krüss DSA100 drop shape analyser.** This droplet is a mass fraction $m_f = 0.0400(3)$ Silica solution, prepared as in Figure 3, resting atop a Parafilm covered glass slide. Note: as the droplet is rotated the contact angle is not necessarily constant due to surface imperfections.

2 Background

Inside crystalline solids, the atoms are fixed in place by a combination of electromagnetic and quantum effects. The total energy required to break the crystal is the sum over all of these interactions over all atoms in the crystal. For example, a Van Der Waal crystal is held together by weak non electrostatic forces which originate from the spatial distribution of electrons around a nucleus:

$$U_{\text{total}} = \frac{1}{2} N 4\epsilon \sum_{i \neq j} \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^{6} \right]$$

where N is the number of atoms in the crystal, and ϵ, r_{ij} are the characteristic energy and discretised nearest neighbour distance respectively. Consider an individual atom inside the body of such a solid (as in Figure 2), the sum over pairwise interactions includes an equal amount of terms in all directions and so the $\frac{1}{r_{ij}}$ terms fall off uniformly with pairwise interactions of atoms inside the crystal. For an atom at the corner or edges of the crystal this spacial symmetry is broken and the energy contribution will include significant terms from atoms in a neighbouring material which are at an interface with the crystal. This interaction can be described by the interfacial energy, which is defined as 'the reversible work per unit area needed to create a new interface' [7]. This concept has been explained using a crystalline solid

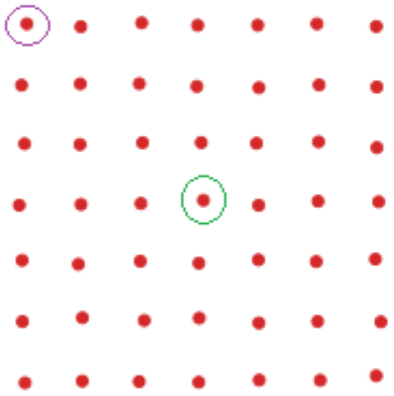


Figure 2: **A finite 2D crystal lattice.** Circled in green is an atom inside the body of the solid; circled in purple is an atom at the surface of the solid. Figure adapted from [8].

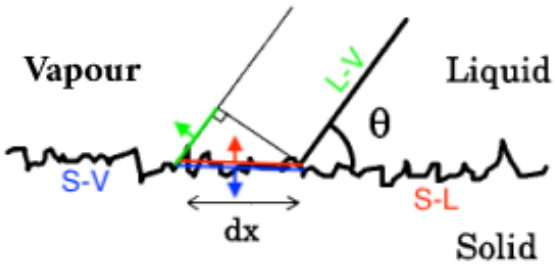


Figure 3: **Infinitesimal displacement (right to left) of the contact line.** Interface boundaries and their respective area elements are colour coded, dA_{S-V} in red, dA_{S-L} in blue and dA_{L-V} in green. dA_{S-V} is decreasing as the contact line shifts from right to left and dA_{S-L} increases by exactly the same amount, indicated by the arrow directions. Figure adapted from [10].

example for simplicity; however, this phenomena also exists for fluids as explained in [9].

2.1 Contact Angles

Now imagine a droplet lying on top of such a crystalline solid, such as in Figure 1, and shifting the three-phase contact line by some small infinitesimal displacement dx parallel to the substrate (Figure 3). The contact angle will subsequently be shifted by $d\theta$ and, assuming the standard thermodynamic coordinates remain unchanged, the surface energy terms will contribute to the free energy:

$$dF = \gamma_{\text{SL}} dA_{\text{SL}} + \gamma_{\text{SV}} dA_{\text{SV}} + \gamma_{\text{LV}} dA_{\text{LV}}$$

where γ are the interfacial tensions for the solid-vapour, solid-liquid and liquid-vapour phase boundaries and dA are the respective area shifts. Consider now that the solid-liquid and solid-vapour interfaces have an equal area shift but opposite normal vector, as argued in [10, 11] and illustrated in Figure 3, so that $dA_{S-V} = -dA_{S-L} = -ldx$ where l is the length of the liquid-vapour boundary. The unit normal vector for dA_{L-V} lies at an angle, θ, with respect to dA_{S-V} and dA_{S-L} so after some algebra:

$$dF = ldx\,(\gamma_{\text{SL}} - \gamma_{\text{SV}} + \gamma_{\text{LV}} \cos(\theta))$$

At equilibrium, one sets $dF = 0$ which obtains:

$$\cos\theta = \frac{\gamma_{\text{SV}} - \gamma_{\text{SL}}}{\gamma_{\text{LV}}} \tag{2}$$

This is called Young's equation. The angle in this relationship is called the contact angle, which is defined as the angle a substance makes with a substrate at the three-phase contact line when the substance is at an equilibrium state. This derivation is not rigorous and a full thermodynamic treatment is detailed in [11]. The fact that the contact angle depends on all three interfacial tensions makes it a good candidate for quantitatively defining a substrate in the laboratory - as is done in this project.

2.2 Feret Diameter

The Feret diameter, which is also commonly referred to as the caliper diameter, will be widely used throughout this project to determine the inner and outer diameters of coffee ring formations. The formal definition of the Feret diameter is the perpendicular distance between two lines tangent to points on the perimeter of an ellipsoid [12] (see Figure 4). In principle, there are an infinite number of Feret diameters as one rotates around an ellipsoid choosing a new set of tangential lines on the ellipsoid's perimeter each time. The soft-

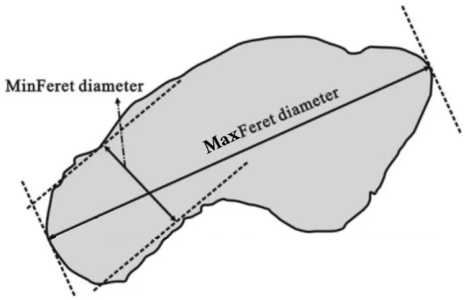


Figure 4: **The Feret Diameter of an ellipsoidal particle.** Figure from [13].

ware used in this project reports only the maximum Feret diameter [14].

Measurement of the Feret diameter on a computer first require the image to be converted to binary. That is, to separate all of the pixels (whose values range from 0-255) in the image into foreground and background pixels. This is done by applying a threshold greyscale value; all of the pixels with a greyscale lower than this are background pixels and the rest are foreground. This process introduces a significant error as parts of the coffee ring will inevitably be determined background when they are actually a part of the foreground (or vice versa) due to variation in greyscale value across the coffee ring perimeter. The uncertainty on a measurement of length for such an image can thus be determined as the standard deviation of greyscale values inside a region of interest on the image. This can be thought of as a measure of how sharp the transition between foreground and background in a given selection is. In this project, the selected region will be the area enclosed by the coffee ring's inner or outer perimeters. Each Feret diameter measurement thus comes with it's own greyscale standard deviation value which can be propagated into individual errors for each ratio $R(\theta)$ measurement (see subsection 4.1).

3 Method

A distilled-water solution of diameter $\phi = 0.3\mu m$ silica particles was prepared by the laboratory manager Dr. Andrew Schofield via the Stober process [15] which were then cleaned and

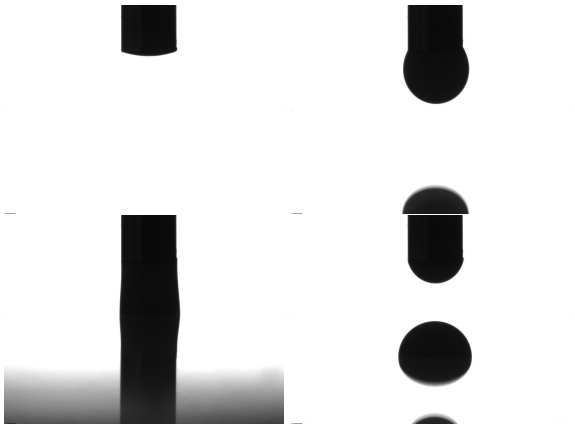


Figure 5: **Pipetting process.** Top-left: glass slide underneath needle. Top-right: $5\mu l$ dispensed from syringe. Bottom-left: platform raised slowly until point of contact. Bottom-right: platform lowered slowly until droplet separates from syringe.

suspended in distilled water (also performed by Dr. Andrew Schofield). The prepared solution had a large mass fraction, $m_f = 0.2546(3)$, which was diluted down into a smaller value of $m_f = 0.0400(3)$ for this project to be dispensed into small $\sim 5\mu l$ droplets (which have a diameter of $\sim 1mm$ on the solid-liquid interface).

3.1 Measurement of the Contact Angle

The silica solution was shaken to ensure even particle dispersion and then loaded into a distilled-water cleaned syringe which was fixed to the Krüss DSA100 drop shape analyser (above the viewing platform). The device used an electric motor to precisely dispense a specified volume of fluid out of the tip of the syringe's needle. Before taking measurements, fluid was manually dispensed until a full droplet exited the needle tip and dropped onto the viewing platform to ensure the needle contained no large air bubbles. A standard volume of $\sim 5\mu l$ was dispensed for each droplet - the precision of this value was not important for this project's aim because the quantity of interest is independent of droplet volume. This value was chosen only because it is small enough so that it does not fall from the needle and large enough so that the droplets obtained on the substrates are large enough to image with the drop shape

analyser.

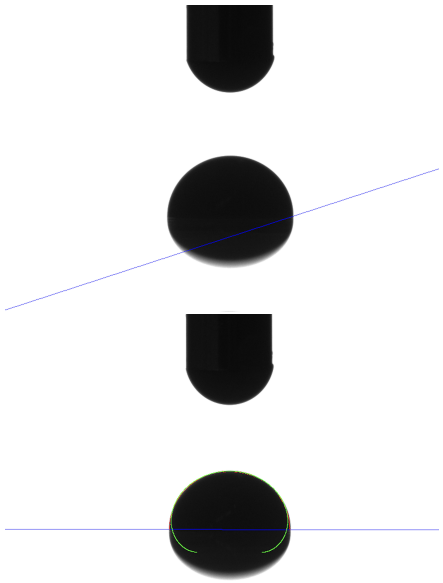


Figure 6: **Examples of fluid-substrate boundary detections.** Top: blue line represents the miscalculated solid-liquid boundary. Bottom: blue line represents an accurate solid-liquid boundary with a green line representing an inaccurate liquid-vapour boundary fit, which leads to an overestimation in the contact angle. These droplets are both resting on ethanol cleaned glass slides.

Next, a substrate was prepared and placed directly below the syringe on the viewing platform. The platform was raised until both the tip of the syringe and the substrate were visible by the camera. The tip of the needle was measured to have a diameter of $\phi = 1.832(1)mm$ using a Schut Electronic Micrometer which was input into the drop shape analyser's software for calibration. A droplet with volume $\sim 5\mu l$ was dispensed and then the viewing platform was raised slowly until the slide *just* touched the droplet. At this point the droplet was forming a bridge between the needle and the substrate. The platform was then lowered slowly until the droplet separated from the needle and began resting on the substrate. This process caused the droplet to vibrate rapidly for a very short time interval which provided many opportunities for the droplet to escape a local minima and reach a global equilibrium state - as per the definition of the contact angle in subsection 2.1. For highly hydrophobic substrates, the platform had to be raised until the needle physically touched the substrate and then lowered until separation. If this was not done the resulting droplet was too small to be analysed by the digital camera of the drop shape analyser. Snapshots of this process are presented in Figure 5.

Before contact angle measurement, the camera's focus was adjusted so that the edges of the droplet appeared sharp. The contact angle measurement process involves the detection of the solid-liquid boundary and the perimeter of the droplet using a digital camera, known as the Sessile Drop technique [16]. The drop shape analyser's automatic solid-liquid boundary detection was used with great success for the highly reflective substrates but manual fitting was required for the less reflective substrates. The baseline fitting was visible on the monitor as measurements were taken - if the baseline was wildly incorrect, such as in Figure 6, then the measurement was discarded.

Ten "good" contact angle measurements were recorded and then the slide was rotated 180° to repeat the process for the other side of the droplet so that surface irregularities are not overlooked [16]. The viewing platform was then adjusted horizontally (perpendicular to the camera and the syringe needle) so that the droplets were separated by $\sim$ 10mm to avoid contamination of the drying surface. This process was repeated for the remaining length of the slide. The slides were transported via an ethanol cleaned Petri dish to a fume cupboard where they were left to dry. The humidity and temperature of the room and the fume cupboard were recorded throughout with a Black Orchid Pro Hydrometer. The influence of these properties are discussed in subsection 5.3.

3.2 Imaging the Ring

After the droplets on a given substrate dried, they were transported, with a closed ethanol cleaned Petri dish, to an Olympus BX50 microscope. The slide was placed on the viewing platform which was then adjusted horizontally and vertically until the sample was visible through the eyepiece. Köhler Illumination steps were conducted with the

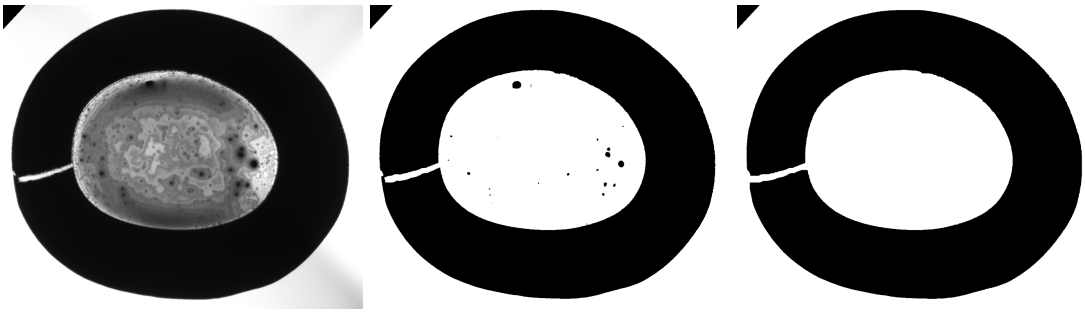


Figure 7: **ImageJ ring analysis process**. Left: unaltered coffee ring on an ethanol cleaned glass slide. Middle: Grey-scale threshold applied and converted image to binary. Right: Image dilated 12 times (white objects expand, black objects shrink) to remove spots inside the ring and then eroded 12 times to recover the ring exactly.

10x focal lens before imaging to maximise the signal to noise ratio on the images, as in [17]. The ring was then imaged with a 5x focal lens starting from the bottom left and going across the rows and then up columns in a snaked fashion. Each image was taken so that approximately 10% of the image overlaps with the preceding image so they can be stitched together easily with the Fiji (ImageJ) software package. To do this consistently, a notable reference point was included between adjacent images, such as a dark spot or a memorable pattern of grey values. This process was repeated for all the droplets on the slide.

3.3 Analysing the Images

The Fiji package collection for ImageJ was used to determine the Feret diameter, area and inner diameter of the ring [14]. First the image was stitched together by saving the .tif image album as a sequence of images. The grid/collection stitching plugin was then used to stitch the images together into one image. Snake by rows and left and up were selected as the image pattern. The ignore Z stage position and subpixel accuracy boxes were ticked and the computation parameters were set to "save computation time (but use more RAM)". The image was then fused and displayed on the screen to be saved in a .one file format.

Next, the stitched image was opened in ImageJ and a threshold adjustment was applied by eye. The lower threshold was adjusted until just before the ring began to visibly erode. The image was then converted to binary so selection tools could be applied. If any spots remained inside the ring, the image was eroded several times to remove them and then dilated an equal amount of times to recover the ring exactly as it was before erosion (see Figure 7). This was to ensure that the wand tool could accurately select the perimeter of the ring. The outer perimeter of the ring was then selected using the wand tool and the binary process was reversed (using the file $\rightarrow$ revert command) to recover the original image, which preserved the grey values of pixels inside the selection. Now a measurement of the Feret diameter was performed and the process was repeated for the inner diameter. If a crack was present in the ring, as in Figure 7, then a manual selection had to be performed using the segmented line tool. The resulting selection was effectively a lower resolution analogue of the wand tool's selection (because it had been selected by hand). Data was then output as a .csv file which was imported to a spreadsheet and python for data analysis.

4 Results

A plot of the obtained data is presented in Figure 8. The Parafilm points situated at $R \approx 2$ and $\theta_c \approx 100^\circ$ are separated from the larger clump of points at $R \approx 1.2$ with the same contact angle. The measurement process for these points did not differ from the rest of the measurements and no human error was recorded so this data cannot be discarded. There are less Parafilm

measurements than for the other substrates because the Parafilm samples were not stretched equally. This was done to attempt to sample a wider range of contact angles; however, the Parafilm was not stretched uniformly which led to very large variances in coffee ring diameters (discussed in more detail in Appendix A). The number of Parafilm points was further reduced by a human error where three droplets on the substrate moved when the slide was picked up, causing them to combine into one big droplet. Whilst the ratio of ring diameter to width is independent of volume, the combination of droplets altered the drying surface underneath the big droplet which invalidates the data point.

However, this does not explain why the presented Parafilm data points are split into two clumps. The measurements were all conducted under the same humidity and temperature so the only viable explanations remaining are that there is either too little data (in which case further experiments would fill in the gap between data points) or some physical process has been overlooked; such as the cleaning of the Parafilm with ethanol or transport between the drop shape analyser room and microscope room.

The Parafilm samples also manifest as two distinct clumps of measurements. The humidity of the room was not controlled and the measurements were conducted at different times where the relative humidity of the room differed by $\sim 20\%$. This result is discussed further in subsection 5.3.

The trend line presented has been computed using the `numpy.polyfit` function to fit the data to a 1D polynomial. This function minimises the least squared error for two given arrays via:

$$E = \sum_i |y(x_i) - x_i|^2 \tag{3}$$

where E is the error to be minimised, x_i are the contact angle measurements and $y(x_i)$ is the interpolated polynomial function.

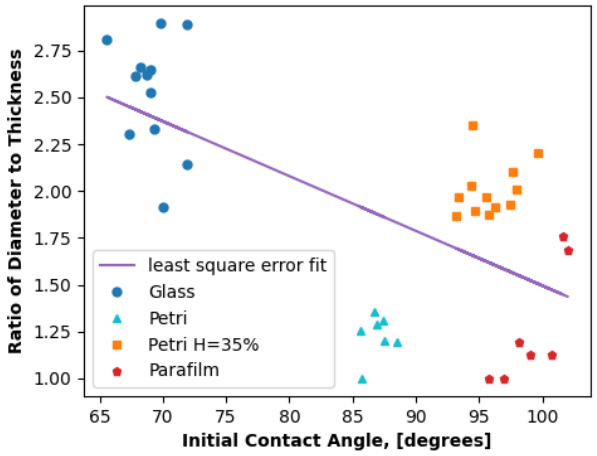


Figure 8: **Ratio of outer diameter to width of various coffee rings.** The purple line is a linear least squares fit, see section 4 for details. Petri samples were conducted on separate days where the relative humidity of the room was not controlled; H represents the recorded relative humidity of the room.

4.1 Error on the Ratio

A plot of $R(\theta)$ against θ with error bars is presented in Figure 9. The uncertainty in the ratio $R(\theta)$ is computed by first considering that each Feret diameter measurement comes with an uncertainty from converting the image to binary using the thresholding tool in ImageJ. The standard deviation of the grey values for a given selection in the image provides a quantitative measure of how precise the perimeter of the ring has been selected (as discussed in subsection 2.2). The ratio is then:

$$R(\theta) = \frac{\phi_{\text{outer}} \pm \delta\phi_{\text{outer}}}{\phi_{\text{outer}} \pm \delta\phi_{\text{outer}} - (\phi_{\text{inner}} \pm \delta\phi_{\text{inner}})}$$

The uncertainties $\delta\phi_{\text{outer}}$ and $\delta\phi_{\text{inner}}$ differ because they were measured using different selections on the image. Out of the 8 plus-minus combinations possible in this expression, the ratio is maximised when the numerator is maximised and the denominator is minimised; likewise, the ratio is minimised when the denominator is maximised and the numerator is minimised. Thus, one defines:

$$R_{\text{max}} = \frac{\phi_{\text{outer}} + \delta\phi_{\text{outer}}}{\Delta\phi - \delta\phi_{\text{outer}} - \delta\phi_{\text{inner}}}$$

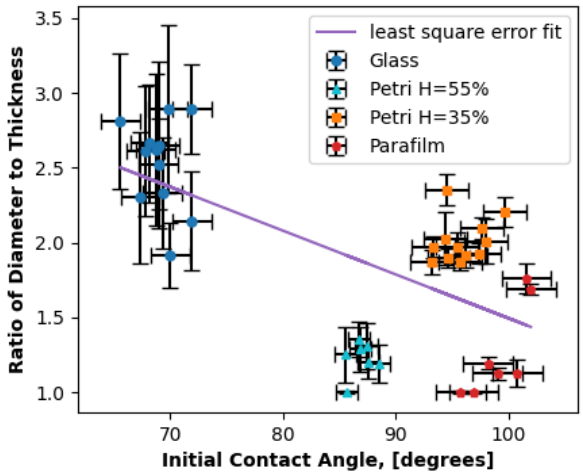


Figure 9: **Ratio of outer diameter to the thickness of ring against contact angle for an ethanol cleaned glass slide.** Uncertainties illustrated with error bars in black.

and

$$R_{\text{min}} = \frac{\phi_{\text{outer}} - \delta\phi_{\text{outer}}}{\Delta\phi + \delta\phi_{\text{outer}} + \delta\phi_{\text{inner}}}$$

where $\Delta\phi = \phi_{\text{outer}} - \phi_{\text{inner}}$ and $\delta\phi > 0$. Then the uncertainty in the ratio is:

$$\delta R = \frac{R_{\text{max}} - R_{\text{min}}}{2}$$

so the quoted values of $R(\theta)$ which appear in Figure 9 are:

$$R(\theta) = \frac{\phi_{\text{outer}}}{\Delta\phi} \pm \frac{R_{\text{max}} - R_{\text{min}}}{2} \tag{4}$$

4.2 Error on the Contact Angle

Contact angle measurements were conducted in groups of 20 per droplet per substrate. The contact angle values presented in this project are the mean values of each set of 20 measurements. The standard error on the mean is proportional to $\frac{1}{\sqrt{20}}$ in this case and is smaller by an order of magnitude than the standard deviation of mean contact angles across a given substrate. Thus, this standard deviation was computed for all of the substrates and is presented as the error bars in Figure 9.

5 Discussion

5.1 Magnitude of Errors

In Figure 8 the different substrates are tightly clumped, with the exception of Parafilm, in both their contact angle values and ratio values which demonstrates homogeneity across the substrates. The error bars for glass are significantly larger than for the other substrates (see Figure 9) on the $R(\theta)$ axis but share a similar order of magnitude of error in contact angle. This asymmetry can be attributed to the less refined microscopy technique used for the glass substrate, as the glass substrate was the first full set of measurements in the project. The other substrates benefit from knowledge of the Köhler Illumination steps; if time was not an issue, the glass measurements could have been repeated with this in mind. For this reason, the glass substrate coffee ring images have saturated white pixel backgrounds (see Figure 7) which leads to large greyscale gradients at the edges of the coffee rings, introducing large Feret diameter measurement errors.

5.2 Trend of the Data

The trend line presented in Figure 8 is a least square error plot as described in section 4. It intersects only a few points and so does not offer any concrete predictive insight. However, the trend does suggest a negative correlation between initial contact angle and ratio of coffee ring diameter to coffee ring width which is in line with measurements in [18]. Thus, a relationship between the initial contact angle and the coffee ring formation on a substrate has been qualitatively confirmed.

5.3 Humidity and Temperature

The conditions of the air in the room appear to affect both the contact angle and coffee ring formation of the droplets, this is supported by the

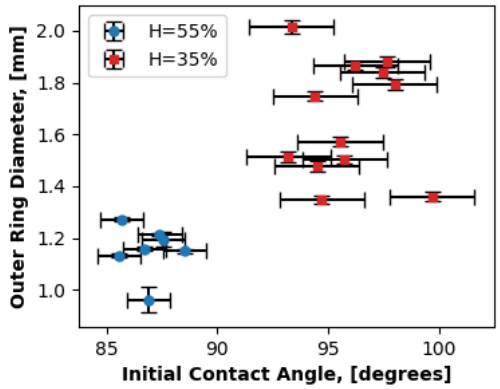


Figure 10: **Contact angle vs outer ring diameter at different relative humidities**. Error bar for the diameter is the greyscale standard deviation value (as discussed in subsection 2.1) converted to mm using calibration slide images. Error on the contact angle is discussed in subsection 4.2

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splitting of Parafilm data present in Figure 8. In the literature it has been shown that the change in evaporation rate of a colloidal droplet due to changes in relative humidity affects both the initial contact angle and the resulting coffee ring particle deposition [19]. The key trends highlighted by Chhasatia et. al. suggest that relative humidity is positively correlated to coffee ring diameter and negatively correlated to the initial contact angle. A plot of initial contact angle vs outer ring diameter using this project's data is presented in Figure 10 which appears to agree with their findings only in part. The samples dried in a $H = 55\%$ environment have a lower contact angle than those at $H = 35\%$ which is in agreement with Chhasatia et. al. but the outer ring diameters appear to display a positive correlation in this case. This phenomenon can likely be attributed to the imprecise volume of the droplets, which is sufficient for comparison of the diameter to width ratio but not for comparison of diameter, where one would expect a larger droplet volume to produce a larger coffee ring diameter. This observation is out of the scope of this project and thus is an avenue for future research.

6 Conclusion

The coffee ring effect was investigated for a silica particle solution (colloid) on a range of substrates with varying hydrophobicity. To do this, an experimental procedure was developed which incorporates a Krüss DSA100 drop shape analyser and an Olympus BX50 microscope to optically measure the initial contact angle of a solution resting on a substrate and then image the coffee ring deposit which is left behind after the droplet has dried. Results show a positive correlation between outer diameter to width ratio and initial contact angle which is consistent with expected trends in [18]. The humidity of the room was overlooked in this project and observations of varying contact angle and outer diameter to width ratio across an identical substrate with two significantly different relative humidity values showed agreement with some of the trends in [19]. Several research avenues have been opened up, including the use of more precise data collection methods to narrow down data clumps and achieve a quantitatively useful trend line and improved microscopy technique to suppress Feret diameter uncertainties

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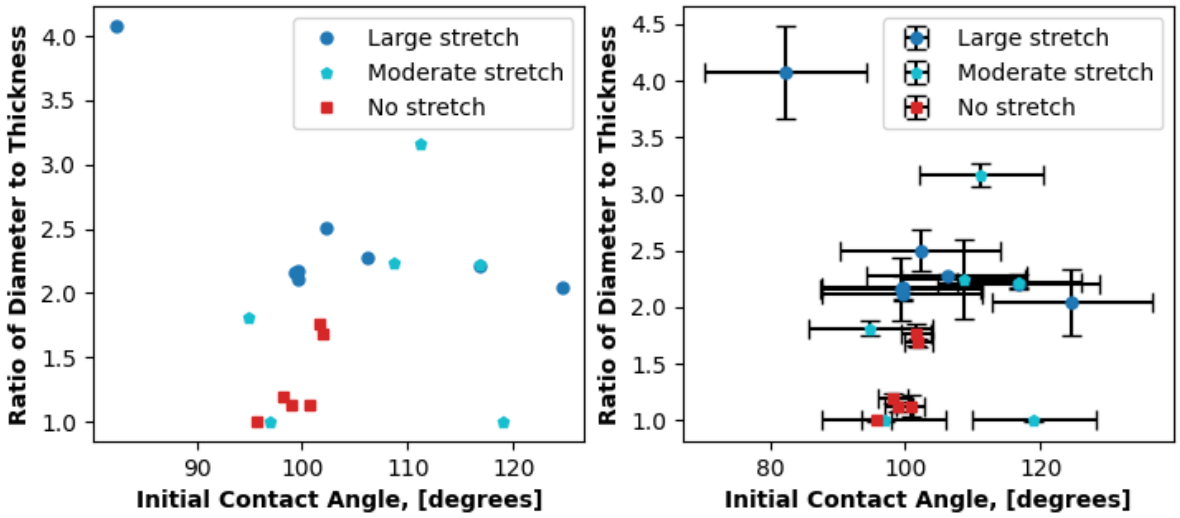


Figure 11: **Contact angle vs $R(\theta)$ for all Parafilm samples.** Error bars presented in a separate plot for clarity. Strong data clumping present for 'no stretch' points and little clumping for the two stretched cases.

Appendix A Degree of Parafilm Stretching

The Parafilm measurements were split into three groups. One slide had an arbitrarily large stretching, one moderately stretched and the third not stretched at all. This was done to sample a larger number of contact angles, with the different stretch factors theoretically acting as new substrates. However, this methodology was severely flawed. The stretching was performed by hand and then the Parafilm was wrapped around an ethanol cleaned glass slide. No measurement was made to quantify the degree of stretching which is the first major pitfall. The stretching being done by hand meant that the surface was not uniformly stretch and thus the solid-liquid and solid-vapour interfacial tensions had large gradients across the surface, the other major pitfall. The stretching is not reproducible and so all of these data points have been removed from Figure 8, but are illustrated in Figure 11. The error bars for both of the stretched cases are significantly larger than for the unstretched case in the θ axis. This indicates that the degree of stretching of the Parafilm substrate is correlated in some way with the variance of contact angle measurements however no definite statements can be made with this data. This stretching effect will not be discussed further in this report, however it has been investigated in detail in [20].